

B.Sc (HONS.) IN CSE FIRST YEAR SECOND SEMESTER EXAMINATION, 2023

DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

Full marks—80

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 V disjunct (+)

Quantile D
 Q3-Q1
 2

[N.B. The figures in the right margin indicate full marks. Answer any four questions.]

Marks

(a) Define proposition, tautology, contradiction, contingency with truth table. 6

(b) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent. 5

(c) Let p and q be the propositions 5

p : You drive over 65 miles per hour

q : You get a speeding ticket

Write these propositions using p and q and logical connectives :

(i) You do not drive over 65 miles per hour.

(ii) You will get a speeding ticket if you drive over 65 miles per hour.

(iii) If you do not drive over 65 miles per hour, then you will not get a speeding ticket.

(d) Find bitwise OR, bitwise AND and bitwise XOR of bit strings 1111 0000, 1010 1010. 4

2. (a) Prove that $\overline{A \cup B} = \bar{A} \cap \bar{B}$. 5

(b) Define disjoint set with example. Suppose that $A = \{1, 2, 4, 5, 6\}$, $B = \{2, 4, 6\}$. Find $B - A$. 5

(c) Use mathematical induction to show that 5

$$a + ar + ar^2 + \dots + ar^n = \frac{ar^{n+1} - a}{r - 1} \text{ when } r \neq 1.$$

(d) Find the Fibonacci numbers f_2, f_3, f_4 and f_5 . 5

[Please turn over]

Marks

(a) What is relation and binary relation with example.

4

(b) Consider the following relation on $\{1, 2, 3, 4\}$

6

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\}$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$$

$$R_6 = \{(3, 4)\}$$

Which of these relations are reflexive, symmetric, antisymmetric and transitive?

(c) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find R^4 .

5

(d) Suppose that the relations R_1 and R_2 on a set A are represented by

5

$$\text{the matrices } M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$.

(a) Define Graph, Multigraph and Pseudograph. Construct a precedence graph for the following program.

6

$$S_1: X = 0$$

$$S_2: X = X + 1$$

$$S_3: Y = 2$$

$$S_4: Z = Y$$

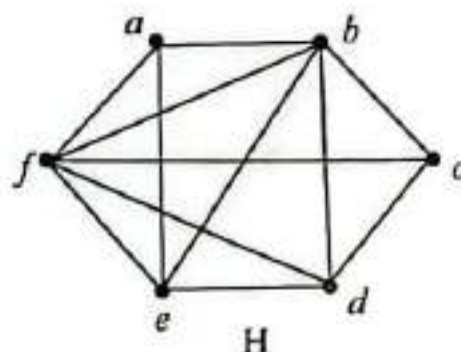
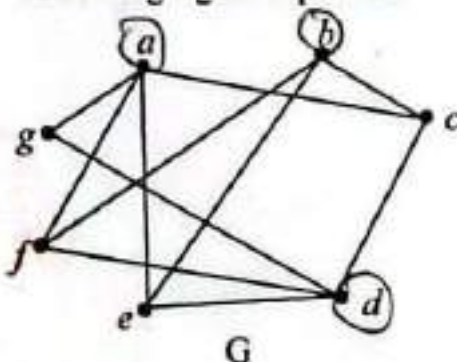
$$S_5: X = X + 2$$

$$S_6: Y = X + Z$$

$$S_7: Z = 4$$

(b) What is bipartite graph? Are the graphs G and H displayed in the following figure bipartite?

5



Incidence matrix

that B equal to the

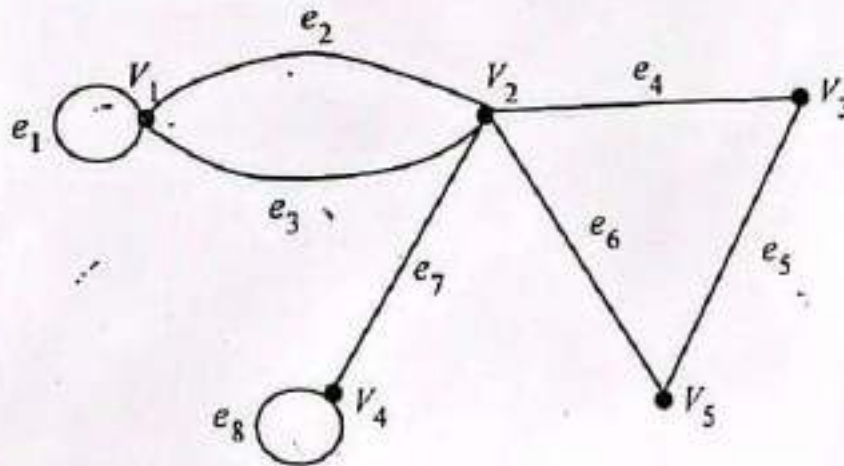
conjugate transpose
of itself

3

Marks

- (c) Represent the pseudograph shown in the following figure using an incidence matrix :

4



- (d) Define planar graph and chromatic number of a graph. What is the chromatic number of K_n ?

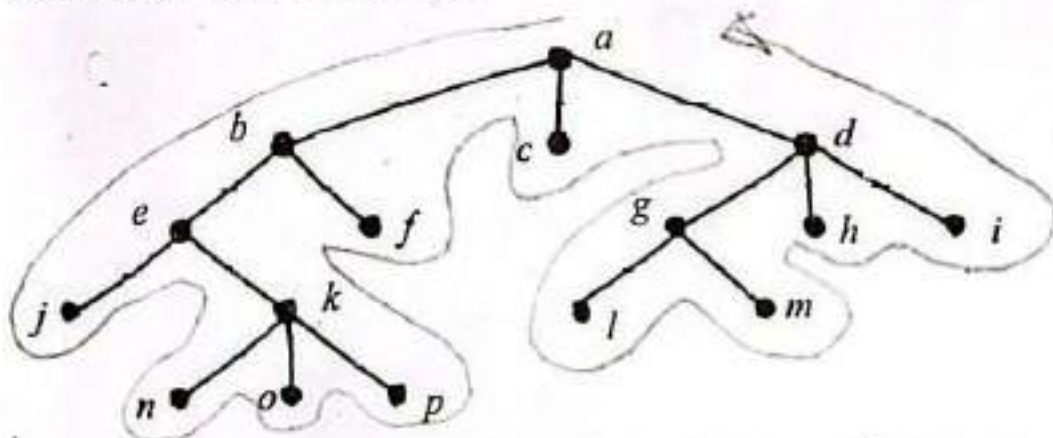
5

- (a) Define rooted tree. Write down the properties of trees.

5

- (b) In which order does a preorder traversal visit the vertices of the following ordered rooted tree :

6



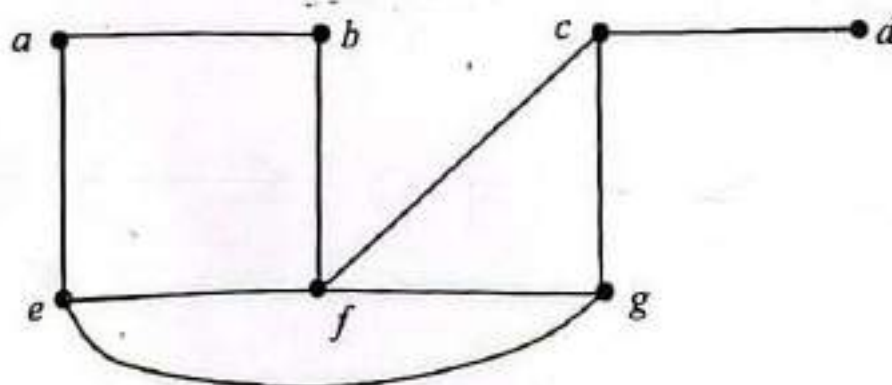
Root left child

- (c) What is the value of the postfix expression $7\ 2\ 3\ * - 4\ \uparrow\ 9\ 3\ /\ +$?

5

- (d) Define spanning tree. Produce three different spanning trees, for the simple graph shown in the following figure :

4



6. (a) Define Boolean expression and Boolean function. Find the values of the Boolean function represented by $F(x, y, z) = xy + \bar{z}$. 5
- (b) What is duality? Find the duals of $x(y + 0)$ and $\bar{x} \cdot 1 + (\bar{y} + z)$. 3
- (c) Construct a circuit for a full adder using two half adders. 6
- (d) Use Karnaugh maps to simplify the following sum-of-product expansions : 6
- (i) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$
- (ii) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$.

$0 = 0$ [more than one solution]
 $0 = 3/4$ [no solution]



$\rightarrow \neg p \leftrightarrow \neg q$

**B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER
EXAMINATION, 2021**

DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : 510223

Examination Code : 5612

Time—3 hours

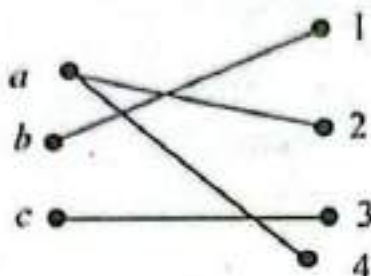
Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any four of the following questions.]

- | | Marks |
|--|-------|
| 1. (a) Which of the following sentences are proposition? What are truth values of those that are propositions?
(i) Savar is the capital of Bangladesh.
(ii) Is he ill?
(iii) $x + 2 = 5$.
(iv) You pass the course. | 4 |
| (b) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements “ x is a lion”, “ x is fierce” and “ x drinks coffee” respectively. Assuming that the universe of discourse is the set of all creatures. Express each of the following statements using quantifiers, logical connectives and $P(x)$, $Q(x)$ and $R(x)$:
(i) All lions are fierce.
(ii) Some lions do not drink coffee.
(ii) Some fierce creatures do not drink coffee. | 8 |
| (c) Consider the following propositions:
p : You can send an E-mail.
q : Your inbox is full.
r : New e-mails are stored in your inbox.
Translate the following sentences into logical expressions:
(i) You cannot send an e-mail if and only if your inbox is full.
(ii) New e-mails are stored in your inbox unless your inbox is full. | 4 |
| (d) Show that $(p \wedge q) \longrightarrow (p \vee q)$ is a tautology. | 4 |
| 2. (a) Define set and power set. What is the power set of $\{a, b, c\}$? | 5 |
| (b) Let $A_i = \{1, 2, 3, \dots, i\}$. Then $\bigcup_{i=1}^n A_i =$ and? $\bigcap_{i=1}^n A_i =$? | 5 |
| (c) Define composition of function. Let f and g be the function from the set of integers to the set of integers defined by the $f(x) = 4x + 2$ and $g(x) = 2x + 1$. What is the composition of f and g ? What is the composition of g and f ? | 5 |

[Please turn over]

- (d) What is function? Is the following figure a function? Why is it or not?

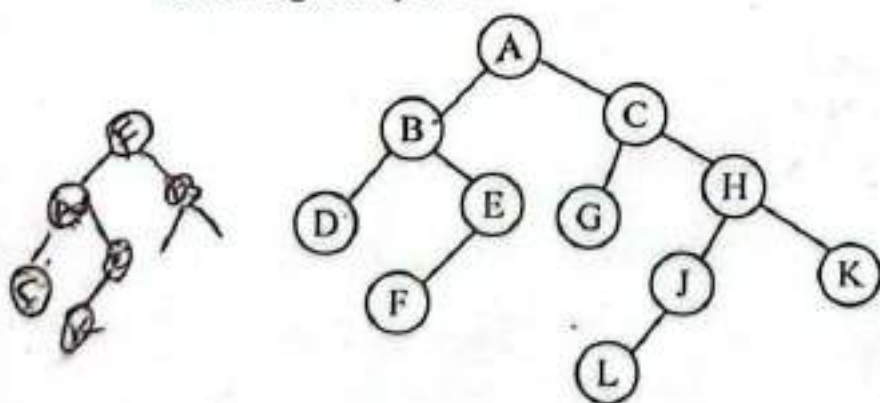


3. (a) What is relation? Write down the properties of relation. 5
 (b) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers $R^n, n = 2, 3, 4, \dots$ 5
 (c) Consider the following relation on $\{1, 2, 3, 4\}$:
 $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1)\}$. Is the relation reflexive, symmetric and transitive? 5
 (d) Find the matrix representation of the relations SoR where the

matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and

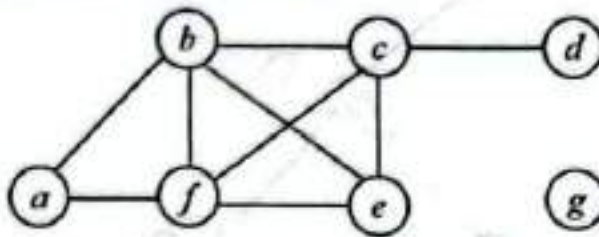
$$M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

4. (a) Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. 5
 (b) Draw a binary tree from the traversal:
 Inorder : E A C K F H D B G 5
 Preorder : F A E K C D H G B
 (c) Write down the preorder, inorder and postorder traversing for the following binary tree : 5



- (d) Evaluate the prefix expression $+ - * 235 / \uparrow 234?$

5. (a) What is graph? What is the degree of vertices in the graph G displayed in following figure : 5



- (b) Draw the precedence graph for the following expression : 5

S1 : $x := 0$

S2 : $x := x + 1$

S3 : $y := 2$

S4 : $z := y$

S5 : $x := x + 2$

S6 : $y := x + z$

S7 : $z := 4$

- (c) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5 ? 5

- (d) Draw a graph with the adjacency matrix : 5

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

6. (a) What is mathematical induction? Using mathematical induction prove that the sum of the first n odd positive integers is n^2 . 5

- (b) Define the Sum Rule and the Product Rule. How many different bit strings are there of length eight? 5

- (c) Describe the universality of NAND and NOR gate. 5

- (d) Simply the Boolean function $F = \overline{ABC} + \overline{BCD} + \overline{ABCD} + \overline{ABC}$ using K-map. 5

**B.Sc. (HONS.) IN COMPUTER SCIENCE AND ENGINEERING
FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020**

DISCRETE MATHEMATICS

[According to the New Syllabus]

Subject Code : CSE-510223

Examination Code : 5612 a Time-3 hours.

Full marks-80

[N.B. The figures in the right margin indicate full marks. Answer any four of the following questions.]

1. (a) Define proposition, tautology, contradiction and conjunction with example. 5
 (b) Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent. 5
 (c) Suppose that $|A| = 4$, $|B| = 7$ and $A \cap B = 3$, Find $A \cup B$. 2
 (d) What is bit string? Find the bitwise XOR for the following strings. 3
 (e) Construct the truth table of the compound proposition $(P \vee \neg Q) \rightarrow (P \wedge Q)$. 5
2. (a) Define rules of inference. Write down the basic rules of inference. 5
 (b) Use mathematical induction to show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$. 5
 (c) Define one to one and onto function. Determine whether the function $f(x) = x^2$ where $f: R \rightarrow R$ It is one to one and onto. 5
 (d) Prove that, $A \subseteq B$, $A \cap B = A$ and $A \cup B = B$ are equivalent. 5
3. (a) Define inverse function and compositions of functions. Use proper diagrams. 5
 (b) What is the truth values of $\exists x P(x)$, where $P(x)$ is the statement $x^2 > 10$ and the universal of discourse consists of the positive integers not exceeding 4. 5
 (c) Describe PASCAL'S IDENTITY and PASCAL'S TRIANGLE with example. 5
 (d) Define equivalence relation. Let m be a positive integer greater than 1. Show that the relation $R = \{(a, b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers. 5
4. (a) Define Cartesian product. What is the Cartesian product $A \times B \times C$, where $A = \{0, 1\}$, $B = \{1, 2\}$ and $C = \{0, 1, 2\}$? 5
 (b) What is the power set of the empty set? What is the power set of the set $\{\varnothing\}$? 5
 (c) Let A and B be two sets of integers and $f: A \rightarrow B$ be a function that represents $f(x) = 4x + 7$. Is f injective or surjective? Give a complete argument supporting your answer. 5
 (d) Prove that $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ where, $a_0, a_1, \dots, a_{n-1}, a_n$, are the real numbers then $f(n)$ is $O(x^n)$.
5. (a) Define tree. Write down the properties of tree. 5
 (b) Prove that, a tree with n vertices has $n-1$ edges. 3

- (c) What is the value of the following postfix notation
 $- 723 \times - 4 \uparrow 93 / + ?$ 3
- (d) Write the pre-order, in-order and post-order traversing for the following tree : 5
- (e) Draw the binary tree from the following traversal : 4
 In-order : KGDLHMBAEC
 Pre-order : ABDGKHL MCE
6. (a) Define graph. How many edges are there in a graph with 10 vertices each of degree six. 5
- (b) State Euler's formula and prove it. 5
- (c) Represent the graph with incidence and adjacency matrix : 5
- (d) Use Dijkstra's Algorithm to find the shortest path between the vertices a and z in the weighted graph shown in figure below:

B.Sc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING FIRST YEAR SECOND SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510223

(Discrete Mathematics)

Time-3 hours

Full marks-80

1. (a) What is proposition? Find the negation of the proposition "Today is Friday" and express this in simple English. 5
- (b) What is tautology? Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology. 5
- (c) Express the statement "Every student in this class has studies calculus" as a universal quantification. 5
- (d) What are the truth values of the propositions $R(1, 2, 3)$ and $R(0, 0, 1)$? 5
2. (a) Define power set. What is the power set of the set $\{0, 1, 2\}$? 4
- (b) Using set builder notation and logical equivalences show that $\overline{A \cap B} = \overline{A} \cup \overline{B}$. 6
- (c) Define one-to-one and onto functions with examples. 5
- (d) Translate the following statements into logical expressions : "Some students in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using quantifiers. 5
3. (a) Define rules of inference. Write down the basic rules of inference. 4
- (b) Using mathematical induction show that, 5
 $1+2+2^2 + \dots + 2^x = 2^{x+1} - 1$ for all nonnegative integers n.
- (c) Define the Sum Rule and the Product Rule. How many different bit strings are there of length seven? 5
- (d) Simplify the sum-of-products expansions using Karnaugh maps : 6
 (i) $xy\bar{z} + \bar{x}\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$.
 (ii) $xyz + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$.
4. (a) Write a recursive procedure for Ackermann function, use the definition of the Ackermann function to find $A(1, 3)$. 5

- (b) Find the matrix representation of the relations SoR where the matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ 5
- (c) How can the final exams at a University be scheduled so that no student has two exams at the same time. [For instance, suppose the courses are numbered 1 to 7 and following pairs of courses have common students :
 (1, 2), (1, 3), (1, 4), (1, 7), (2, 3), (2, 4), (2, 5), (2, 7), (3, 4), (3, 6), (3, 7), (4, 5), (4, 6), (5, 6), (5, 7) and (6, 7)]. 5
- (d) If G is a connected planar simple graph with e edges and v vertices, $v \geq 3$ then $e \leq 3v - 6$. 5
5. (a) Define multigraph and pseudograph with examples. 5
 (b) Construct BST of the following values : 5
 14, 3, 4, 12, 11, 5, 2, 8, 2, 7, 9, 16, 6, 20.
 (c) Draw the Hasse diagram representing the partial ordering $\{(a, b) ; a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. 5
 (c) Evaluate the prefix expression $+ - * 235 / ^ 234$. 5
6. (a) State Euler's theorem and prove it using required diagram. 5
 (b) Describe the universality of NAND & NOR gate. 5
 (c) What is the chromatic number of the Graph G and H : 5
 (d) Determine the order in which a pre-order traversal visits the vertices of the following ordered rooted tree ; 5

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2018

[According to the New Syllabus]

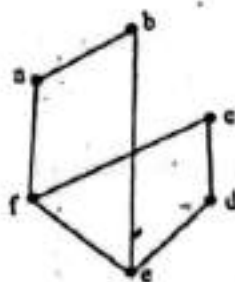
CSE-510210

Examination Code: 5612

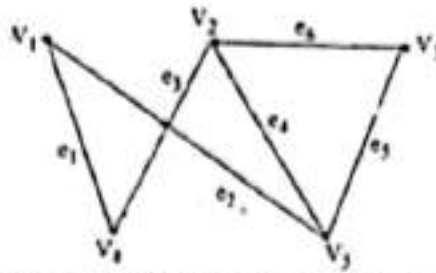
(Discrete Mathematics)

1. (a) Define Cardinality of set, Infinite set and Power set. What is the power set of the set $\{1, 2, 3\}$? 3+2=5
 (b) Let $A = \{i, i+1, i+2, \dots\}$. Then $\bigcup_{i=1}^n A_i = ?$ and $\bigcap_{i=1}^n A_i = ?$ 5
 (c) Let A and B be two sets of integers and $f: A \rightarrow B$ be a function that represents $f(x) = 4x + 7$. Is f injective or surjective? 5
 (d) Compute the following sum : $S = \sum_{k=50}^{100} K^2$. 5
2. (a) What is predicate and quantifiers? If $Q(x, y)$ denotes the statement ' $x = y + 3$ ' What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$? 5
 (b) What is the complement of a set? Let A, B and C be sets, show that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$. 5
 (c) What is duality? Find the duals of $x(y + 0)$ and $\bar{x}.1 + (\bar{y} + Z)$. 5
 (d) Show different types of correspondences of function by figure: 5

- (i) One-to-one, not onto
 (ii) Onto, not one-to-one
 (iii) One-to-one, and onto
 (iv) Neither one-to-one nor onto
 (v) Not a function
3. (a) What is mathematical induction? Using mathematical induction prove that the sum of the first n odd positive integers is n^2 . 5
 (b) Give an indirect proof of the theorem "If $3n + 2$ is odd, then n is odd". 4
 (c) Define sum rule and product rule. How many bit strings of length eight either start with a 1 bit or end with the two bits 00? 6
 (d) State pigeonhole principle. What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D and F? 5
4. (a) What is relation? Write down the properties of relation. 4
 (b) Consider the following relations on the set of integers : 6
 $R_1 = \{(a, b) | A \leq b\}$
 $R_2 = \{(a, b) | A > b\}$
 $R_3 = \{(a, b) | a = b \text{ or } a = -b\}$
 $R_4 = \{(a, b) | a = b\}$
 $R_5 = \{(a, b) | a = b + 1\}$
 $R_6 = \{(a, b) | a + b \leq 3\}$
 Which of these relations contain each of the pairs $(1, 1)$, $(1, 2)$, $(2, 1)$, $(1, -1)$ and $(2, 2)$?
 (c) Define partial ordering. Construct a Hasse diagram for $(\{1, 2, 3, 4\}, \leq)$. 6
 (d) Draw the directed graph for the relation : 4
 $R = \{(1, 1), (1, 3), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), 4, t)\}$.
5. (a) Define multigraph and pseudograph. Is K_4 planar graph or not? $2+3=5$
 (b) Is the graph shown in figure below bipartite? Justify your answer. 5



- (c) Represent the graph shown in the following figure with an incidence matrix. 5

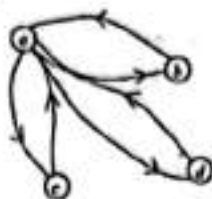


- (d) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5 ? 5
6. (a) Define Boolean expression and Boolean function. 3
- (b) Prove the absorption law $x(x + t) = x$ using the other identities of Boolean algebra. 5
- (c) Construct a Full Adder circuit using Half Adder circuits. 6
- (d) Simply the Boolean function $F = \overline{A}BC + \overline{B}CD + \overline{A}BCD + \overline{A}BC$ using K-map. 6

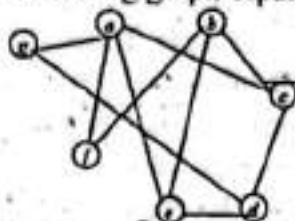
B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER
EXAMINATION, 2017
CSE-127
(Discrete Mathematics)
Examination Code : 612

1. (a) Define proposition. What are the contra positive, the converse and the inverse of the implication "If I am elected then I will lower taxes"? 1+3=4
- (b) Let p, q and r be the propositions—
p : You have the flu
q : You miss the final examination
r : You pass the course
Express each of the following propositions as the English sentence:
(i) $p \vee q \vee r$; (ii) $(p \wedge q) \vee (\neg p \wedge r)$. 5
- (c) Show that, $\neg(p \leftrightarrow q)$ and $\neg p \leftrightarrow q$ are logically equivalent. 6
- (d) Show that, $p \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$. 5
2. (a) What is function? Let f and g be the function from the set of integers to the set of integers defined by the $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of g and f? What is the composition of f and g? 5
- (b) Use set builder notation and logical equivalences to establish the first De Morgan law $\overline{A \cap B} = \overline{A} \cup \overline{B}$. 5
- (c) Show that $\sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r - 1}$, where $r \neq 1$. 5
- (d) Use mathematical induction to show that $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$. 5
3. (a) Define Relations. Exemplify the most important properties of relations. 5
- (b) Use the mathematical induction to show that $1 + 2 + 2^2 + 3^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n. 5
- (c) Let $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$. Find the powers R^n , $n = 2, 3, 4, \dots$. 5

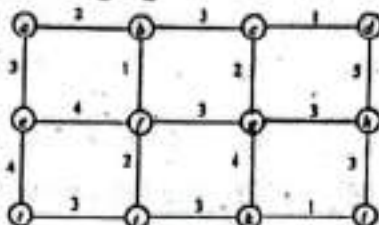
- (d) Determine whether the relation for the following directed graph is reflexive, symmetric, antisymmetric and transitive: 5



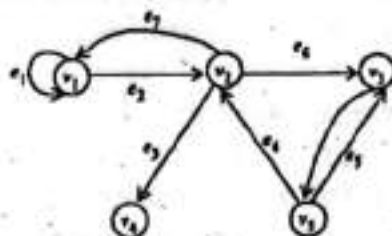
4. (a) Define graph. How many edges are there in a graph with ten vertices each of degree six? 1+3=4
(b) Define bipartite graph. Is the following graph bipartite? 5



- (c) What is minimum spanning tree? Use Prim's algorithm to find a minimum spanning tree in the graph shown in the following figure:



- (d) Define adjacency and incidence matrix. Represent the pseudo, graph shown into the following figure using an incidence matrix:



5. (a) State Euler's formula and prove it
(b) Define binary search tree. Using alphabetical order, construct a binary search tree for the words in the following sentence:—
The quick brown fox jumps over the lazy dog.
(c) What is the prefix form for $((x + y) \uparrow 2) + ((x - 4)13)$?
(d) Use K-map to minimize the sum-of-product expansion
 $xyz + xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$
6. (a) Construct full adder using half adder. 5
(b) What is the value of the following postfix notation? 5
 $723 \times 4 \uparrow 9 3 / + ?$
(c) Describe the universality of NAND and NOR gate. 5
(d) Draw the Hasse diagram representing the partial ordering $\{(a, b) | a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$. 5

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2016**CSE-.127****(Discrete Mathematics)****Examination Code : 612**

1. (a) Consider the following propositions :- 3
 - p : you can send an e-mail.
 - q : your inbox is full:
 - r : new e-mails are stored in your inbox.

Translate the following sentences into logical expressions:----

 - (i) you cannot send an email if and only if your inbox is full.
 - (ii) new e-mails are stored in your inbox unless your inbox is full.
- (b) What do you mean that two propositions, are equivalent? 6

Prove that, $\neg p \vee q$ and $p \rightarrow q$ are equivalent
- (c) Define, converse, contra positive and inverse 5
- (d) Translate the following statements to logical expressions 6

All hummingbirds are richly colored." "No large birds live on honey."
 "Birds that do not live on honey are dull in color." "Hummingbirds are small."
2. (a) Define function. Is the function $f(x) = x^2$ from the set of integers 3

one-to-one correspondence?
- (b) Prove that the integer n is odd if and only if n^2 is odd. 6
- (c) Use mathematical induction to prove that the sum of the first n 6

odd positive integers is n^2
- (d) State pigeonhole principle. What is the minimum number of 5

students required in a discrete mathematics class, to be sure that at least six will received the same grade if there are five possible grades A, B, C, D, and F?
3. (a) Consider the following relation on $\{1, 2, 3, 4\}$:- .

$R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1)\}$.

Is the relation reflexive, symmetric and transitive?
- (b) What is the composition of the relations R and S , where R is 4

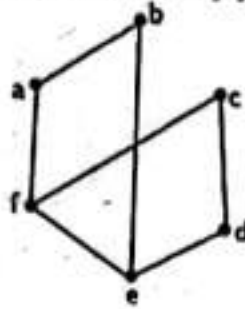
the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?
- (c) Define partial ordering. Construct the Hasse diagram for $(\{1, 2, 3, 4\}, \leq)$. 6
- (d) Let m be a positive integer -greater than, 1. Show that the 5

relation $R = \{(a, b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers.
4. (a) How many bit strings of length eight start with "1" or end with the "00"? 5
- (b) What is the coefficient of $x^{25}y^{25}$ in the expansion of $(x + y)^{25}$? 5
- (c) How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where $x_1 + x_2 + x_3$ are non negative integers? 5
- (d) In how many ways can 12 students be partitioned into four 5

teams, so that each team contains three students?
5. (a) Define graph, multi graph and pseudo graph. 3

(b) Prove that an undirected graph has even number of vertices of odd degree. 5

(c) Is the graph shown in fig below bipartite? Justify your answer. 6

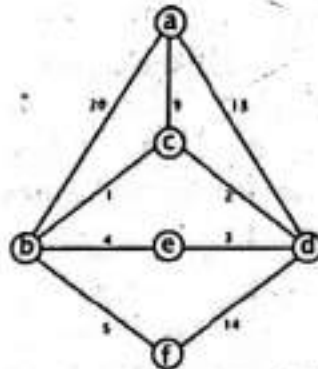


(d) Draw the graphs: 6

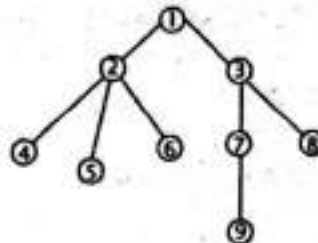
(i) $K_{3,5}$ (ii) Q_3 (iii) W_5 (iv) K_5

6. (a) Define tree. Prove that a tree with n vertices has $n-1$ edges. 5

(b) Use Kruskal's Algorithm to find the minimum cost spanning tree from the following graph : 5



(c) Determine the order in which a pre-order traversal visits the vertices of the given ordered rooted tree 5



(d) Use Karnaugh maps to simplify the SOP expressions- 5

(i) $xy_z + xyz$:

(ii) $xyz + xyz + syz + xyz + xyz$

B.Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINATION, 2015

CSE-127

Examination Code : 612

(Discrete Mathematics)

Time-3 hours

Full marks—80

1. (a) Define proposition, conjunction, implication, tautology and contradiction. 5

(b) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent. 5

(c) Use mathematical induction to prove that the sum of the first n odd 5

positive integers is n^2 .

(d) What is quantification? Describe different types of quantifier.

1+4=5

2. (a) Define one-to-one and onto functions with examples.

5

(b) Is the following figure a function? Why is it or not?

4

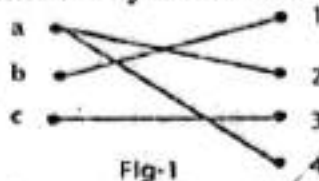


Fig-1

(c) Give an indirect proof of the theorem "If $3n + 2$ is odd, then n is odd."

4

(d) Use mathematical induction show that-

$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n .

(C) Give a recursive definition of $\sum_{k=0}^n ak$

2

3. (a) Write down the properties of relations.

5

(b) Define composite relation. Consider the following relation on $\{1, 2, 3, 4\}$: $1+5=6$ $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$.

Is the relation reflexive, symmetric and transitive?

(c) State two basic counting principle.

5

(d) Suppose that $|A| = 4$, $|B| = 7$ and $|A \cap B| = 3$. Find $|A \cup B|$.

4

4. (a) Define a simple graph. What are the degrees of the vertices in the graph displayed in the following figure?

5

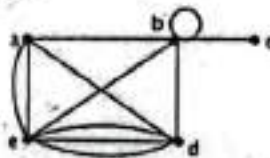


Fig-2

(b) Construct a precedence graph for the following program :- 5

$S_1 : a := 0$

$S_2 : b := 1$

$S_3 : c := a + 1$

$S_3 : d := b + a$

$S_5 : e := d + 1$

$S_6 : e := c + d$

(c) Represent the graph shown in the following figure with an incidence matrix.

5

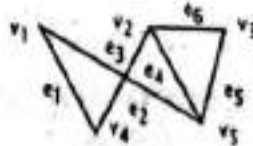


Fig-3

(d) What is planar graph? Is the following graph shown in figure planar? write in details.

5

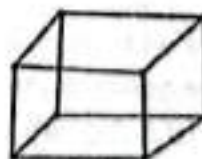
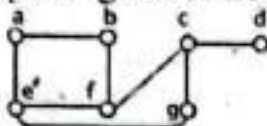


Fig-4

- (a) Represent the expression $((x + y) T2) + ((x - 4)/3)$ using binary tree. 4
 (b) Write down the preorder, in order and post order traversing for the following tree. 6



- (c) What is Spanning tree? Find a spanning tree of the following simple graph G : 3



- (d) Write down the different properties of tree. 5
 6. (a) What is Boolean function of degree n? Find the values of the Boolean function represented by $F(x, y, z) = xy + z$ 5
 (b) Define-duality. Find the duals of $x(y + 0)$ and $x.1 + (y' + z)$. 4
 (c) Construct a Full Adder circuit using two Half Adder circuits. 5
 Use Karnaugh maps simplify the sum-of-products expansions 6
 (i) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$:
 (ii) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$.

B.Sc (HON&) IN CSE PART I; SECOND SEMESTER EXAMINATION, 2014

CSE-127 Examination Code : 612

(Discrete' Mathematics)

Tim-3 hours

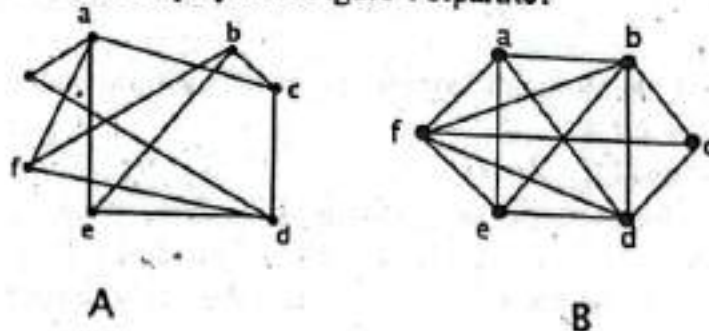
Full marks---80

1. (a) show that $1(p \vee (1p \wedge q))$ and $1p \wedge 1q$ are logically equivalent [without using truth table]. 5
 (b) Write down the De Morgan's Laws for quantifiers with when true and when false 5
 (c) Let $A_i = \{i, i + 1, i + 2, \dots\}$
 Then, $\bigcup_{i=1}^n A_i = ?$ and $\bigcap_{i=1}^n A_i = ?$ 5
 (d) Define Inverse function and compositions of function. 2+3=5
 Suppose that $|A| = 10, |B| = 5$ and $|A \cap B| = 2$ Find $|A \cup B| = ?$
 2. (a) Define function. Is the function $f(x) = x^2$ from the -set of integers to the set of integers one-to-one correspondence? 3
 (b) Give a big-O estimate, for $f(x) = 3n \log(n!) + (n^2 + 13) \log n$, where n is a positive integer. 5
 (c) Give a direct proof of the theorem "if n is odd, then n^2 is odd." 6
 (d) Use mathematical induction -to prove that the sum of the first n odd: 6 positive integers is n^2 . 2
 3. (a) Define Inclusion-Exclusion Principle with example. 6
 (b) Determine whether the sequence (a_n) is a solution' of the Recurrence relation $a_n = f_n = 2a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$, where $a_n = 3n$ for every non-negative integer n. Answer the same question

where $a_n = 3n$ and where $a_n = 127$.

- (c) Write down the sum and product theorem for generating function. 6
- (d) Show that Extended binomial co-efficient $C(-n, k) = (-1)^k C(n+r-1, r)$.
4. (a) Define relation. Describe the properties of relation. 5
- (b) What is the composite of relations R and S where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?
- (c) Define partial ordering. Construct Hasse diagram for the poset $(\{1, 2, 3, 4\}, S)$. 5
- (d) Let M be a positive integer greater than 1. Show that the relation $R = \{(a, b) \mid a \equiv b \pmod{M}\}$ is an equivalence relation on the set of integers. 4
5. (a) Define graph, multi graph and pseudo graph. 3
- (b) Prove that an undirected graph has an even number of vertices of odd degree. 6
- (c) Construct a precedence graph for the following program :- 6
- $S_1: x := 0$
- $S_2: x := x + 1$
- $S_3: y := 2$
- $S_4: z := y$
- $S_5: x := x + 2$
- $S_6: y := x + z$
- $S_7: z := 4$

(d) Are the graphs G and H displayed in Figure 1 bipartite?



6. (a) Write down the truth table and logic circuit for AND, OR, NOT, Ex-OR and Ex-NOR gates. 5
- (b) Describe the universality of NAND and NOR gate. 5
- (c) Find Boolean product from the following matrices :- 5
- $$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$
- (d) Construct Half Adder circuit with truth table. 5

B.Sc (HONS) IN CSE PART-I, SECOND SEMESTER EXAMINATION-2013

CSE-127

(Discrete Mathematics)

Examination Code: 612

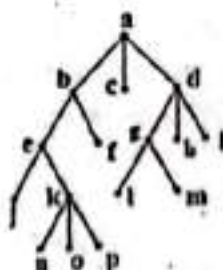
1. (a) Define Conjunction, Disjunction, Implication and Contra positive with truth table. 5
 (b) Define relation and function. What are the properties of relation? 5
 (c) Let $A = \{0, 2, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$ and $C = \{0, 3, 6, 9\}$, What are the $A \cap B \cap C$ and $A \cup B \cup C$? (Using Venn diagram and normal set theory.)
 (d) Show that $\neg(p \wedge (\neg p \wedge q))$ and $\neg q \wedge \neg q$ are logically equivalent. 5
2. (a) Define rules of inference. Write down the basic rules of inference. 6
 (b) Define mathematical induction. Why mathematical induction is a valid proof technique? 6
 (c) Prove the theorem "The integer n is odd if and only if n^2 is odd." 5
 (d) Consider two set $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 3, 5, 7, 9\}$. 5
 (i) Find the bit strings of A and B .
 (ii) Use bit string to find the union and intersection of these sets.
3. (a) Suppose a graph represented by the following table:- 5
 $G = [x : y, z, w; y : x, y, w; z : z, w; w : z]$
 (i) Find the number of vertices and edges in G .
 (ii) Are there any sources or sinks?
 (iii) Draw the graph of G .
- (b) Let A, B, C, D, E, F, G, H be eight data items with the following assigned weights:- 8

Data	A	B	C	D	E	F	G	H
item:								
Weight	22	5	11	19	2	11	25	5

Construct a 2-tree T with a minimum weighted path length p using the above data as external nodes.

- (c) Construct the graph $G = \{1, 2, 3, 4, 5, 6\}$ under multiplication modulo 7. 7
 (i) Find the multiplication table of G .
 (ii) Is G cyclic?
 (iii) Find $2^{-1}, 3^{-1}, 6^{-1}$.
 (iv) Find the orders and sub groups generated by 2 and 3

4. (a) Define chromatic number. What is the chromatic number of $K_{3,4}$ and K_5 ?
 (b) What is the limitation of binary search? Write the binary search tree algorithm.
 (c) What do you mean by tree traversing? Travers the following tree into post order:-



- (d) What is the value of the post fix expression $723 * -4 \uparrow 93 / + ?$ 2
 (e) Write the procedure of preorder algorithm. 3

5. (a) Use mathematical induction to show that, $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integer n . 6
- (b) Prove that $\sqrt{2}$ is irrational number by giving a proof by contradiction.
- (c). State and prove the Euler's theorem $V - E + K2$. 6
6. (a) What is the co-efficient of $x^{12}y^{13}$ in expansion of $(2x-3y)^{25}$? 5
- (b) How many one-to-one functions are there from a set with m elements to one with n elements? 8
- (c) Show that $f_n > \alpha^{n-2}$, where $\alpha = (1+\sqrt{5})^2$, when ever $n \geq 3$. 3
- (d) Show that the assertion "All primes are odd" is false.

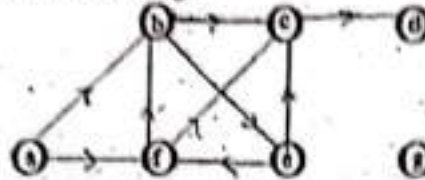
B.Sc (HONS) IN CSE PART-I, SECOND SEMESTER EXAMINATION-2012
CSE-127

(Discrete Mathematics)

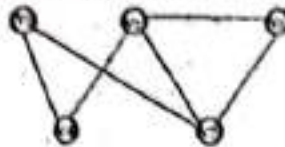
1. (a) Define Proposition, Negation, Conjunction, Implications and Biconditional with example.
- (b) Which of the following sentences are proposition? What are truth values of those that are propositions?
- (i) Comilla is the capital of Bangladesh.
- (ii) Are you sick?
- (iii) $x+1=3$.
- (iv) What time is it?
- (v) $2+2=3$.
- (c) Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$, where A and B where $A=\{1, 2\}$ and $B=\{a, b, c\}$.
- (d) Define Inverse function and Compositions of functions. Let f and g be the function from the set of integers to define by the $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition off and g ? What is the composition of g and f ?
2. (a) Define mathematical induction. Use mathematical induction to prove that, $n > 2^n$ for all positive integers n .
- (b) Give an indirect proof of the theorem "If $3n+2$ is odd, then n is 4 odd".
- (c) Define floor and ceiling function. Let a and b be positive integers, and Q is a function recursively defined as follows:
- $$Q(a, b) = \begin{cases} 0 & \text{if } a < b \\ Q(a - b, b) + 1 & \text{if } a \geq b \end{cases}$$
- Find (i) $Q(2, 5)$ (ii) $Q(12, 5)$.
- (d) Define Fibonacci sequence. Find the Fibonacci numbers f_2, f_3, f_4, f_5 . 5
3. (a) Define rules of inference. Write down the basic rules of inference. 6
- (b) Define mathematical induction. Why mathematical induction is a valid proof technique?
- (c) Use mathematical induction to show that, $1 + 2 + 2 + \dots + 2n = 2^{n+1} - 1$.
- (d) Give a big O estimate for $f(n) = 3n \log(n!) + (n^2 \log 3) \log n$, where n is a positive integer.
4. (a) Define graph, multi graph and pseudo graph with example.
- (b) Draw the precedence graph for the following expression:
- $$S_1: x := 0$$
- $$S_2: x := x + 1$$
- $$S_3: y := 2$$

- $S_4 : z : y$
 $S_5 : x : x + 2$
 $S_6 : y : x + z$
 $S_7 : z : 4$

- (c) Define in-degree and out-degree. What are the in-degree and out-degree of the vertices in the graph G displayed in following:-



- (d) Define incidence matrices. Use an incidence matrices to represent the following graph:-



- (e) Draw a graph with the adjacency matrix:

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

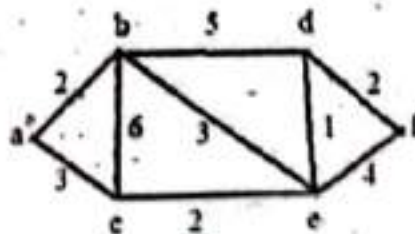
5. (a) State two basic counting principles.
 (b) How many bit string of length eight either start with a 1 bit or end with the bit 00?
 (c) What is the composite of the relations R and S, where R is the relation from {1, 2, 3} to {1, 2, 3, 4} with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from {1, 2, 3, 4} to {0, 1, 2} with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?
 (d) Define binary relation. Consider the following relations on {1, 2, 3, 4}
 $R = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$
 Is the relation reflexive, symmetric and transitive?
6. (a) Define Boolean expression and Boolean function. How many different Boolean functions of degree are there?

- (b) Let A and B be two Boolean Matrices given as follows:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Find the Boolean product of A and B that is A B.

- (c) Define spanning tree. Using Prim's Algorithm find the minimum cost spanning tree from the following figure:-



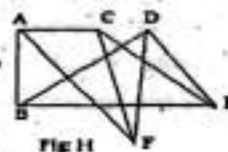
- (d) What is the values of the postfix expression $\pm - * 2 3 5 / 1 2 3 4$?

B.Sc (HONS) IN CSE PART-I, SECOND SEMESTER EXAMINATION-2011

CSE-127

(Discrete Mathematics)

1. (a) Define proposition, disjunction conjunction, tautology and 5 contra-positive.
(b) Which of the sentence are proposition? Write true or false with 5 cause:
 (i) Birds can fly;
 (ii) Answer the question;
 (iii) $2 + 35$;
 (iv) $x + 2 = 11$;
 (v) $5 + 7 = 1$
 (c) Let $A = \{0, 2, 4, 6, 8\}$, $B = \{0, 1, 2, 3, 4\}$ and $C = \{0, 3, 6, 9\}$. What 5 are $A \cap B \cap C$ and $A \cup B \cup C$? (Using Venn diagram and normal set theory).
 (d) Define relation and function. Write the properties of relation. 5
2. (a) Define Handshaking Theorem. Discuss graph representation technique in memory. 5
 (b) What is 'the chromatic number of C_6 ? Write the necessary and sufficient conditions for Euler Circuits and path.
 (c) Prove that 'in an undirected graph even number of vertices of odd degree.
 (d) What is planar graph? Are $K_{3,3}$ and Q planar?
3. (a) i) Prove the theorem "The integer n is odd if and only if n^2 is odd". 4+3=7
 ii) Write the properties of isomorphic graph.
 (b) Are the graph (G) is isomorphic with (H) graph explain it.



- (c) Are the above graph (G) and (H) bipartite or not? 4
 (d) State two basic counting principles and pigeonhole principle. 5
4. (a) What is the composite of relation R and S where R is the 5 relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to $\{0, 1, 2\}$ with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$?
 (b) How many bit strings of length eight start with a 1 bit or end 5 with the 2 bits 00?
 (c) What do you mean by graph coloring? Write down it's some applications.
 (d) What do you mean by SOP and POS? Find the SOP expansion for the following function:
 $f(x, y, z) = (x + y) \bar{z}$
5. (a) Define mathematical induction. Why mathematical induction is valid proof technique?
 (b) Use mathematical induction show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n .
 (c) Define one-to-one and onto function. Determine whether the 5 function $f(x) = x^2$ from the set of integers to the set of integers is one-to —one and onto.
 (d) What is spanning tree? Show that a simple graph is connected if 5 it has a spanning tree.
6. (a) Define Boolean expression and Boolean function: Find the sum of-products expansion for the function $f(x, y, z) = (x + y)$.
 (b) Construct a Half adder using logic gates.
 (c) Prove the absorption law $x(x+t) = x$, using the other identities of Boolean algebra.

- (d) Use karnaugh maps simplify the sum-of-products expressions:

$$\overline{xyz} + x\overline{y}z + \overline{x}yz + \overline{xy}z + \overline{xyz}$$

**B.Sc(HONS.) IN CSE(PART-I), SECOND SEMESTER
EXAMINATION-2010**

CSE-127

(Discrete Mathematics)

- 1 (a) Define proposition, tautology and contradiction with example. 4
- (b) Show the $\neg(p \vee (\neg p \wedge q)) \neg p \wedge \neg q$ logically equivalent. 5
- (c) Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a lion", "x is fierce" and "x drinks coffee", respectively. Assuming that the universe of discourse is the set of all creatures. Express each of the following statements using quantifiers, logical connectives and $P(x)$, $Q(x)$ and $R(x)$: 8
 - (i) All lion are fierce;
 - (ii) Some lions do not drink coffee;
 - (iii) Some fierce creatures do not drink coffee;
 - (iv) Does (iii) from (i) and (ii), If not, is there a correct conclusion?
- (d) Use set builder notation and logical equivalences to show that 3

$$\overline{A \cap B} = \overline{A} \cap \overline{B}.$$
- 2 (a) Define rules of inference. Write down the basic rules of inference. 6
- (b) Give an indirect proof of the thermo "If $3n+2$ is odd, then n is odd". 5
- (c) Define mathematical induction. Why mathematical induction is a valid proof technique? 4
- (d) Use mathematical induction to show that 5

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1.$$
- 3 (a) State two basic counting principles. 5
- (b) How many bit string of length eight either start with a bit or end with the two bit 00? 5
- (c) What is the 5 composite of the relation R and S , where R is the relation from $\{1, 2, 3\}$ to $\{1, 2, 3, 4\}$ with $R = \{(1, 1), (1, 4), (2, 3), (3, 1), (3, 4)\}$ and S is the relation from $\{1, 2, 3, 4\}$ to with $S = \{(1, 0), (2, 0), (3, 1), (3, 2), (4, 1)\}$. 5
- (d) Define binary relation. Consider the following relation on $\{1, 2, 3, 4\}$: 5

$$\{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}.$$
 Is the relation reflexive, symmetric and transitive?
- 4 (a) Define Multigraph, Directed graph and Pseudograph with figures. 3
- (b) Construct a precedence graph for the following statements:- 4

$$S_1: x = 0$$

$$S_2: x = x + 1$$

$$S_3: y = 2$$

$$S_4: z = y$$

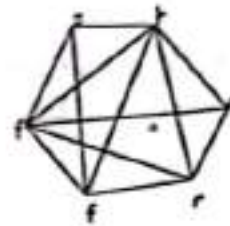
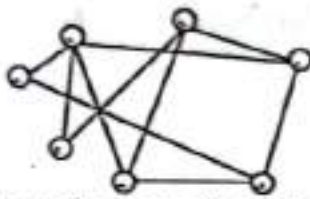
$$S_5: x = x + 2$$

$$S_6: y = x + z$$

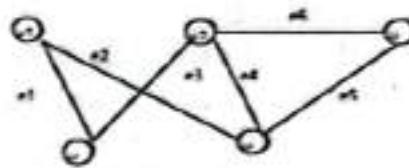
$$S_7: z = 4$$

- (c) Define bipartite graph. Are the graph and displayed in the following figure bipartite? Explain.

6



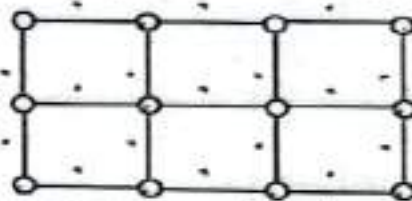
- (d) Prove that an undirected graph has an even number of vertices of odd degree. 3
 (e) What is incidence matrices? Construct an incidence matrix for the undirected graph shown in the following figure:- 4



- 5 (a) Define m -ary tree and binary tree. Prove that a full m -ary tree with i internal vertices contains $n = mi + 1$ vertices. 5
 (b) Find the pre-order traverse for the following tree:- 5



- (c) What is the post fix form of the expression $((x+y) \uparrow 2) + ((x-4)/3)?$ 3
 (d) What is minimum spanning tree? Use Prim's algorithm to find a minimum spanning tree the graph shown in the following figure: 7



- 6 (a) Define Boolean expression and Boolean function. How many different Boolean functions of degree n are there? 6
 (b) Find the sum of products expression for the function $F(x, y, z) = (x + y) \bar{z}$ 4
 (c) Using Karnaugh maps to simplify the sum of products expressions $xy\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}\bar{z}$. 5
 (d) *Construct a full adder using logic gates. 5